

VIJAY WANKHEDE

Senior Full Stack Engineer · Fintech, Payments & Web3

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PROFESSIONAL SUMMARY

Senior full stack engineer with 6+ years designing, scaling, and operating high-throughput distributed systems for fintech, payments, and Web3 products. Primary language TypeScript (NestJS, React, fp-ts) with strong secondary Python (FastAPI) for services, data, and tooling.

Deep experience with event-driven architectures, Postgres and Redis, and cloud-native delivery on Kubernetes (EKS/GKE). Built production ledger, routing, and settlement systems with idempotent APIs, exactly-once semantics, and first-class observability. Comfortable in low-structure, high-ambiguity environments — ship to learn, own the outcome, reason from first principles.

Actively integrate AI-driven development (Claude Code, Cursor, Codex) and AI-assisted testing into the delivery loop to move faster without trading reliability.

CORE COMPETENCIES

Full Stack Engineering (TypeScript + Python) · React & Data-Dense UIs · High-Throughput Distributed Systems · Ledger, Routing & Settlement · Event-Driven Architecture · Idempotency, Retries & Eventual Consistency · Observability (Metrics, Tracing, Logging) · Postgres & Redis · Kubernetes on GCP/AWS · Web3 / L2s / Smart Contracts · Zero-Knowledge Proofs · Fraud & Anomaly Detection · AI-Driven Development · Low-Ego, High-Ownership Delivery

TECHNICAL SKILLS

Languages: TypeScript, Python, Solidity (familiar)

Frontend: React, Next.js, TypeScript, Redux / Zustand, real-time UIs

Backend: Node.js / NestJS, Express, fp-ts (functional TypeScript), FastAPI, Flask, gRPC

Architecture: Microservices, Event-Driven, CQRS, Hexagonal, REST / GraphQL / gRPC / WebSockets, idempotency, retry/backoff, eventual consistency

Messaging / Streaming: Apache Kafka, Redis Streams, RabbitMQ, AWS SQS / SNS / EventBridge, GCP Pub/Sub

Databases & Caches: PostgreSQL (partitioning, replication, logical decoding), Redis (cache, pub/sub, locks, streams), DynamoDB, MongoDB

Observability: Prometheus, Grafana, OpenTelemetry, Sentry

Web3 & Privacy: Ethereum / L2s (Arbitrum, Optimism, Base), smart contracts, EIP-2771 meta-tx, zkSNARKs for privacy-preserving audit trails

Testing & Quality: Jest, Vitest, Pytest, Playwright, integration & contract tests, Artillery load testing

AI-Driven Development: Claude Code, Cursor, Codex

PROFESSIONAL EXPERIENCE

Catalysis Network — Staff Full Stack Engineer

Remote · 2025 – Present

- Lead full stack design and delivery across a **TypeScript**-first stack — **NestJS** services, a **React** data-dense operator UI, and **Python / FastAPI** services for data, risk, and internal tooling — for a platform launching with \$1.5M+ in user assets.
- Architected event-driven microservices on **Kubernetes** (AWS EKS, portable to GKE) with **Kafka**, **PostgreSQL**, and **Redis**, using fp-ts-style Either/Task error handling to enforce correctness across the ledger, router, and settlement services.

- Owned the Router middleware end-to-end: load-balanced services behind an API gateway, idempotent write paths, optimistic concurrency, and replay-safe Kafka consumers — guaranteeing exactly-once semantics for restake, reward, and slashing flows under sustained load.
- Shipped a signature-based settlement flow using **EIP-2771 meta-transactions** (off-chain signer, TypeScript validation pipeline, on-chain adapter) — eliminating user friction in the consumer path while preserving full cryptographic auditability for a regulated, high-audit environment.
- Built the React operator dashboard: real-time status of validator sets, reward flows, and settlement queues over WebSockets, with server-state managed via React Query and optimistic UI for sub-second perceived latency.
- Stood up observability from zero — **Prometheus** metrics, **Grafana** dashboards, OpenTelemetry traces, structured JSON logs, Sentry alerting, and 99.9% SLOs on critical write paths — and authored the on-call runbooks the team uses today.
- Drove engineering velocity by integrating **Claude Code**, **Cursor**, and **Codex** into the day-to-day loop: built custom Claude skills for codegen, architectural review, and adversarial test generation; paired Codex with **CodiumAI / Qodo** and **Meticulous** for AI-assisted unit, integration, and visual regression testing — cutting feature lead time on the StakeManager service by ~40% with no regressions in production.
- Mentored engineers, ran weekly design reviews, and raised the bar through structured PR review and shared TypeScript / functional-programming standards.

Synthr Protocol — Senior Full Stack Engineer

Remote · 2023 – 2025

- Led the full stack for a production cross-chain transfer / payments product across Axelar and LayerZero — **TypeScript** relay, event indexer, and off-chain verification layer, plus REST / GraphQL APIs and the React partner console powering consumer-facing flows.
- Re-architected the message pipeline around **Kafka** with consumer groups, dead-letter queues, and exactly-once semantics, lifting customer retention 30% by eliminating stuck transactions and exposing real-time status to end users.
- Designed and shipped a NestJS event indexer ingesting logs from 6+ chains into **PostgreSQL** with reorg handling, backfill, and a **Redis**-backed cursor store for resumable consumption — production-grade under bursty load.
- Built an anomaly-detection service in **Python** over live bridge transaction streams — graph features plus isolation-forest / simple z-score rules — to flag suspected fraud and replay loops before settlement, feeding into an operator-facing React review queue.
- Stood up Grafana + Prometheus monitoring from scratch, defined SLIs/SLOs for relay latency and indexer lag, and built alerting that paged on-call before users noticed degradation.
- Set TypeScript coding standards, reviewed every backend PR, and onboarded three engineers across the relay and indexer codebases.

Instabit — Backend Engineer (Payments & Fintech)

Remote · 2021 – 2023

- Built a production payments confirmation pipeline in **Python (FastAPI)** and Go handling thousands of transaction events per minute, with idempotent processing, mempool monitoring, and configurable confirmation thresholds per merchant.
- Designed the merchant-facing REST API and webhook delivery system, including HMAC-signed payloads, exponential-backoff retries, and a dead-letter queue for poison messages — patterns I now treat as table stakes for any payments integration.
- Migrated the core ledger from a single-node **PostgreSQL** instance to a replicated, partitioned setup with read replicas, cutting p99 read latency ~60% and unblocking customer-facing reporting.
- Containerized services with Docker, deployed to AWS, set up GitHub Actions CI/CD with automated migrations, and introduced structured logging that became the basis for the team's incident response process.

Deqode Labs — *Software Engineer*

Indore, India · 2019 – 2021

- Built **TypeScript / Node.js** backend services on NestJS for fintech and supply-chain clients — JWT auth with refresh rotation, RBAC, audit logging, and high-volume event listeners ingesting on-chain data into PostgreSQL.
- Contributed to **React** admin dashboards and wrote integration test suites with Jest and Supertest, raising backend coverage on the flagship project from ~35% to ~80%.
- Collaborated directly with clients on API design, system trade-offs, and deployment planning — early experience that shaped how I run technical design reviews today.

EDUCATION

B.E., Computer Engineering — Institute of Engineering and Technology, India